Jansky's discovery, showed that, if the then-current ideas about how cosmic radio waves might be generated were correct, it was *impossible* for our Milky Way galaxy to produce radio waves as strong as Jansky was seeing. Despite this apparent impossibility, Greenstein and Whipple believed Jansky's observations; they were sure the problem lay with astrophysical theory, not with Jansky. But with no hints as to where the theory was going wrong, and since, as Greenstein recalls, "I never met anybody else [in the 1930s] who had any interest in the subject, not one astronomer," they turned their attention elsewhere.

By 1935 (about the time that Zwicky was inventing the concept of a neutron star; Chapter 5), Jansky had learned everything about the galactic hiss that his primitive antenna would allow him to discover. In a quest to learn more, he proposed to Bell Telephone Laboratories the construction of the world's first real radio telescope: a huge metal bowl, 100 feet (30 meters) in diameter, which would reflect incoming radio waves up to a radio antenna and receiver in much the same way that an optical reflecting telescope reflects light from its mirror up to an eyepiece or a photographic plate. The Bell bureaucracy rejected the proposal; there was no profit in it. Jansky, ever the good employee, acquiesced. He abandoned his study of the sky, and in the shadow of the approach of World War 11, turned his efforts toward radio-wave communication at shorter wavelengths.

So uninterested were professional scientists in Jansky's discovery that the only person to build a radio telescope during the next decade was Grote Reber, an eccentric bachelor and ham radio operator in Wheaton, Illinois, call number W9GFZ. Having read of Jansky's radio hiss in the magazine *Popular Astronomy,* Reber set out to study its details. Reber had a very poor education in science, but that was unimportant. What mattered was his good training in engineering and his strong practical streak. Using enormous ingenuity and his own modest savings, he designed and constructed with his own hands, in his mother's backyard, the world's first radio telescope, a 30-foot (that is, 9-meter)diameter dish (Figure 9.1c); and with it, he made radio maps of the sky (Figure 9.1d). In his maps one can see clearly not only the central region of our Milky Way galaxy, but also two other radio sources, later called Cyg A and Cas A—*A* for the "brightest radio sources," *Cyg* and *Cas* for "in the constellations Cygnus and Cassiopeia." Four decades of detective work would ultimately show, with high probability, that Cyg A